A GPU search for large prime gaps in primorial windows, with a new record merit

Mario Vojvoda (independent researcher, Samobor, Croatia) · abstract prepared for arXiv endorsement (math.NT), September 2026

Abstract. We describe BEACON, a search for large gaps between consecutive primes in windows around $n_0(k) = k \cdot 1439\#/2310$ (611-digit integers: the primorial of 1439 with the primes 2, 3, 5, 7, 11 divided out). Candidates in each window are sieved on a GPU to prime bound 10^{10} and tested by Fermat tests arranged as a ladder of concentric rings, so that a window is abandoned as soon as a prime appears on either side; a lookup-table predictor ranks windows by their full-width sparsity before testing. Windows that pass most rings but fail an outer one are re-examined by a walk that locates the true first primes on both sides; this walk recovers the asymmetric gaps that a symmetric ring test discards, and it produced the two largest gaps found. Every reported gap was independently re-derived from (k, j_l, j_r): all interior integers proven composite (sieve factor or failed base-2 Fermat test), both endpoints proven prime by ECPP (CM 0.4.3), every certificate checked with PARI/GP. The search found a prime gap of 59,724 following the prime $17440361647223 \cdot 1439\#/2310 - 21464$, of merit 42.5192, exceeding the previously largest known merit (41.9388, 2017); a second gap of 58,416 (merit 41.5882) ranks second. Twenty gaps from the search were accepted into the Prime Gap List on 22 September 2026; about 23,000 further first-occurrence improvements (merit 19–36), certified the same way, await bulk entry. We give the throughput model of the pipeline (a 4-GPU workstation tests about 44,000 windows per second), measured ring-pass rates, and the cost of the search.

Principal results (accepted C?D in the Prime Gap List, 22 Sep 2026; ECPP certificates for all endpoints)

gap	merit	start prime	digits
59,724	42.5192	17440361647223 * 1439#/2310 - 21464	611
58,416	41.5882	17325055497313 * 1439#/2310 - 29514	611
54,184	38.5755	17216776002509 * 1439#/2310 - 13782	611
52,980	37.7186	17039226586301 * 1439#/2310 - 27198	611
52,868	37.6945	2144030080399 * 1439#/2310 - 34758	610
... 15 more	35.8 - 37.3	same form	610-611

Verification status

Interior: every integer between the endpoints proven composite on two machines (sieve to 10^9, base-2 Fermat on survivors; 0 pseudoprimes). Endpoints: ECPP certificates from CM 0.4.3 (Enge), each verified independently with PARI/GP *primecertisvalid*. The Prime Gap List server (primegaps.cloudygo.com) re-verified endpoints and interiors on acceptance. Certificates and reconstruction tuples are deposited with the submission.

Content of the paper

1. Windows, the swap-prime family and the choice of 1439#/2310. 2. The GPU sieve (bucketed records for primes to 10^{10}) and its identity gates against a CPU reference. 3. The ring ladder: expected primes per ring, measured pass rates (agreement with design to 1% over $2.8 \cdot 10^{10}$ windows), the early-exit schedule. 4. Sparsity prediction: residue tables for primes to 10^5, correlation −0.40 with true survivor counts, ×1.42 gain per tested window. 5. The salvage walk and the asymmetric-gap channel. 6. Verification protocol and certificates. 7. Throughput, cost, and expected time to a given merit. Code (CUDA/C/Python) is released with the paper. Related work: Troisi, *Combined Sieve Algorithm for Prime Gaps*, arXiv:2012.03771; Nicely's first-occurrence and maximal gap tables; the Prime Gap List project.